MILITARY SPECIFICATION

FUZES, GENERAL PURPOSE GRENADE: M230 and M231 (PRACTICE)

This Amendment forms a part of Military Specification MIL-F-63446A (AR), dated 12 September 1983, and is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1.2, Delete "Fuze Assembly-M230 All ways" and substitute "Fuze General Purpose Grenade: M230".

Delete "Fuze Assembly-M231 All Ways" and substitute "Fuze, General Purpose Grenade, Practice: M231".

PAGE 5

3.5.7, Delete "18 ± 1.5 pounds" and substitute "18 ± 2 pounds".

3.5.8, Delete "two pounds" and substitute "five pounds".

3.5.9, Delete ".25 seconds" and substitute ".25 seconds (M230 Fuze) or 1.75 ± .75 (M231 Fuze)".

PAGE 6

Add new requirement 3.8:

"3.8 Arming pin/slider lock. The arming pin shall lock the slider in its fully safe position when the fuze is fully assembled."

PAGE 7

4.3.1, Delete "Fuze Subassembly No. 1", "Fuze Subassembly No. 2", "Fuze Subassembly No. 4", "Fuze Subassembly No. 5", and "Fuze Subassembly No. 6" in their entirety.

Delete "Fuze Subassembly No. 3" and substitute "Fuze Body Subassembly".

FSC 1330

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PAGE 8

4.3.1, Delete "Fuze Assembly, M231 Allways" and substitute "Fuze, General Purpose Grenade, Practice: M231".

Delete "Fuze Assembly, M230 Allways" and substitute "Fuze General Purpose Grenade: M230".

Add under Submission "Drive Screw, Modified 9390620 25".

PAGE 10

TABLE I, Delete "Fuze Subassembly No. 1" and "Fuze Subassembly No. 2" in their entirety.

Delete "Fuze Subassembly No. 3" and substitute "Fuze Body Subassembly".

Delete "Pushout requirement" and substitute "Anti-twist pin pushout requirement".

Add "sleeve pushout requirement 25 3.4 Gage".

PAGE 11

TABLE I, Delete "Fuze Subassembly No. 4" in its entirety.

PAGE 15

TABLE I, Delete "Fuze Subassembly No. 5" in its entirety.

PAGE 16

TABLE I, Delete "Fuze Subassembly No. 6" in its entirety.

PAGE 17

TABLE I, Add "Drive Screw, Modified
(Dwg 9390620)
Examination for Defects 25 3.4 4.4.2.52"

PAGE 18

TABLE I, Delete "M231 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade, Practice: M231".

Add: "Torque Requirement 115 (a) .40% 3.4 4.4.2.50".

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Delete: "Transportation Vibration 20 3.5.3 4.5.8/4.5.15/4.5.16" and substitute "Transportation Vibration 20 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c)".

Delete "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15/4.5.16" and substitute "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete "Thermal Shock 30 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 30 3.5.6 4.5.11/ 4.5.15 (c)/4.5.16 (c)".

Delete "Arming Time 115 (a) 3.5.9 4.5.14 (b)".

Delete "Fuzing No-Go 115 (a) 3.5.10 4.5.15 (b)" and susbtitute "Fuzing No-Go 49 (d) 3.5.10 4.5.15 (b)".

Delete "Fuzing Go 115 (a) 3.5.11 4.5.16 (b)" and substitute "Fuzing Go 66 (d) 3.5.11 4.5.16 (e)".

In Note (b), Delete "32" and substitute "16" in two places; Delete "51" and substitute "17".

Add to the Notes Block "(c) Fuzing No-Go - 1/2 test sample per temperature; Fuzing Go - 1/2 test sample per temperature. (d) Selected from fuzes marked (a). (e) $50^{o} \pm 5^{o}F$ -22 Fuzes; $70^{o}F \pm 15^{o}F$ -22 Fuzes; $160^{o}F \pm 5^{o}F$ -22 Fuzes".

PAGE 19

TABLE I, Delete "M230 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade, M230".

Add "Torque Requirement 115 (a) 3.4 4.4.2.51".

Delete "Transportation Vibration 20 3.5.3 4.5.8/4.5.15 4.5.16" and substitute "Transportation Vibration 20 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c)".

Delete "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15/ 4.5.16" and substitute "Helicopter Vibration 30 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete "Thermal Shock 30 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 30 3.5.6 4.5.11/4.5.15 (c)/ 4.5.16 (c)".

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Delete "Arming Time 115 (a) 3.5.9 4.5.14 (b)" and substitute "Arming Time 115 (a) 3.5.9 4.5.14 (f)".

Delete "Fuzing No-Go 115 (a) 3.5.10 4.5.15 (b)" and substitute "Fuzing No-Go 49 (d) 3.5.10 4.5.15 (b)".

Delete "Fuzing Go 115 (a) 3.5.11 4.5.16 (b)" and substitute "Fuzing Go 66 (d) 3.5.11 4.5.16 (e)".

In Note (b), Delete "32" and substitute "16" in two places; Delete "51" and substitute "17".

Add to the Notes Block "(c) Fuzing No-Go - 1/2 test sample per temperature; Fuzing Go - 1/2 test sample per temperature. (d) Selected from fuzes marked (a). (e) $50^{\circ} \pm 5^{\circ}F$ -22 Fuzes; $70^{\circ}F \pm 15^{\circ}F$ -22 Fuzes; $160^{\circ}F \pm 5^{\circ}F$ -22 Fuzes". (f) Tests to be performed at cold, ambient, and hot temperatures in the quantities described in (b) and (e)."

PAGE 21

4.4.2.1, Major 102: Delete "width of slider" and substitute "width of slider rear".

Major 103: Delete "thickness of slider" and substitute "thickness of slider rear".

* Critical: Delete "None defined".

Add:

"Critical 1: Presence of red lacquer on bore rider 100% 3.4 Visual".

"Major 117: Minimum coverage of red lacquer on bore rider .40% 3.2 Gage".

"Major 118: Width of bore rider tip .40% 3.4 Gage".

"Major 119: Thickness of slider front .40% 3.4 Gage".

"Minor 209: Maximum coverage of red lacquer on bore rider .65% 3.4 Gage".

"Minor 210: Width of slider between ends .65% 3.4 Gage".

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"Minor 211: Thickness of slider between ends .65% 3.4 Gage".

PAGE 24

4.4.2.2, add Major 129: Diameter of Drive Screw Holes .40% 3.4 Gage".

PAGE 28

4.4.2.5, Delete in its entirety.

PAGE 30

4.4.2.7, Delete in its entirety.

PAGE 32

4.4.2.9, Delete "Fuze Subassembly No. 3" and substitute "Fuze Body Assembly".

Delete "9347186" and substitute "9333825 or 9345168".

* Add Majors 102 and 103 as follows:

"102 Pins above bottom of slider slot .40% 3.4 Gage
103 Anti-twist pin pushout requirement .40% 3.4 Gage"

Delete Minors "201", "202", and "203" in their entirety.

Minor 204: Delete "Pushout requirement" and substitute "Sleeve pushout requirement".

PAGE 33

4.4.2.10: Delete in its entirety.

PAGE 34

4.4.2.11, Add Major 111: "Finish on tip (a) 3.4 Gage".

Add to Notes block: "(a) Special Inspection Level 4 of MIL-STD-105D."

PAGE 38

4.4.2.15, Minor 203: Delete in its entirety.

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4.4.2.32: Delete "9347187" and substitute "9333825".

PAGE 56

4.4.2.33: Delete in its entirety.

PAGE 58

4.4.2.35, Add "Major 104: Tip length 100% 3.4 Gage".

PAGE 59

4.4.2.36, Major 101: Delete "Tip length" and substitute "Straight pin recess".

Delete "100%" and substitute ".40%".

PAGE 62

4.4.2.39, Major 101: Delete "100%" and substitute ".40%".

Major 102: Delete "wire" and substitute "clip".

PAGE 64

4.4.2.41: Delete in its entirety.

PAGE 73

4.4.2.50: Delete "M231 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade, Practice: M231".

Delete Major 104 in its entirety.

Add Critical 9, 10, and 11:

"9	Safety pin missing	100%	3.4	Visual
10	Fuzing No-Go	100%	3.5.10	4.5.15
11	Arming pin/slider lock	100%	3.8	4.5.15"

Add Majors 106 and 107:

"106	Excessive adhesive	100%	3.4	Visual
107	Torque requirement	.40%	3.4	Gage"

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Delete "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 4.5.16" and substitute "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c)".

Delete "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15/ 4.5.16" and substitute "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete "Thermal Shock 12 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 12 3.5.6 4.5.11/4.5.15 (c)/ 4.5.16 (c)".

Delete "Shear Wire Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Shear Wire Pull Test 64 (a) 3.5.7 4.5.12".

Delete "Safety Pin Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Safety Pin Pull Test 64 (a) 3.5.7 4.5.12".

Delete Major 103 in its entirety.

Major 104, delete "(a)" and substitute "(d)".

Major 105, delete "(a)" and substitute "(d)".

Add to Notes block "(c) Fuzing No Go - 1/2 test sample per temperature; fuzing Go - 1/2 test sample per temperature. (d) Selected from fuzes marked (a)."

PAGE 74

4.4.2.51: Delete "M230 All Ways Fuze Assembly" and substitute "Fuze, General Purpose Grenade: M230".

Delete Major 104 in its entirety.

4.4.2.51: Add Critical 9, 10, and 11:

"9	Safety pin missing	100%	3.4	Visual
10	Fuzing No-Go	100%	3.5.10	4.5.15
11	Arming pin/slider lock	100%	3.8	4.5.15"

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Add Majors 106 and 107:

"106	Excessive adhesive	100%	3.4	Visual
107	Torque requirement	.40%	3.4	Gage"

Delete "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 4.5.16" and substitute "Transportation Vibration 12 3.5.3 4.5.8/4.5.15 (c)/4.5.16 (c)".

Delete "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15/ 4.5.16" and substitute "Helicopter Vibration 12 3.5.4 4.5.9/4.5.15 (c)/4.5.16 (c)".

Delete "Thermal Shock 12 3.5.6 4.5.11/4.5.15/4.5.16" and substitute "Thermal Shock 12 3.5.6 4.5.11/4.5.15 (c)/ 4.5.16 (c)".

Delete "Shear Wire Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Shear Wire Pull Test 64 (a) 3.5.7 4.5.12".

Delete "Safety Pin Pull Test 32 (a) 3.5.7 4.5.12" and substitute "Safety Pin Pull Test 64 (a) 3.5.7 4.5.12".

Delete "Arming time 32 (a) 3.5.9 4.5.14 (b)" and substitute "Arming time 62 (a) 3.5.9 4.5.14 (b)".

Major 104, delete "(a)" and substitute "(d)".

Major 105, delete "(a)" and substitute "(d)".

Add to Notes block "(c) Fuzing No Go - 1/2 test sample per temperature; fuzing Go - 1/2 test sample per temperature. (d) Selected from fuzes marked (a)."

4.4.2.52, add included Page (74a).

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4.5.9: Delete "MIL-STD-810C" and substitute "MIL-STD-810B".

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER 9390620
4.4.2.52	Screw, Modified Drive				NEXT HIGHER ASSEMBLY 9333825 or 9345168
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
CRITICAL	None defined.				
MAJOR 101	Thread Body Diameter		0.40%	3.4	Gage
MINOR 201	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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4.5.11: Delete "The fuzes shall not require special preparation prior to this test" and substitute "Each fuze shall be sealed in a temperature stable plastic bag prior to the test."

4.5.12: Delete "less than 16.5 pounds or more than 19.5 pounds" and substitute "less than 16 pounds or more than 20 pounds".

PAGE 81

4.5.13: Delete "less than 1 pound or greater than 2 pounds" and substitute "less than 1 pound or greater than 5 pounds".

4.5.14: Delete ".25 seconds" and substitute ".25 seconds (M230 Fuze) or 0.75 ± .11 second (M231 Fuze)".

4.5.15, delete in its entirety and substitute:

"4.5.15 Fuzing No-Go and arming pin/slider lock. Each fuze shall be subjected to and pass the Fuzing No-Go and the Arming Pin/Slider Lock Test.

WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming, application of impulse, and re-safing. Test equipment design and detailed operating and testing procedures must be approved by ARDC Safety Office.

The fuze shall be secured in a holding fixture in such a manner that the shock impulse that the fuze senses will not be attenuated or magnified beyond that applied. The design of the fixture shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat.

After mounting the fuze to the holding fixture, but prior to performing the test, the safety pin and shear wire shall be removed. Removal of either the safety pin or shear wire shall be considered as beginning of the arming operation and thus safety shielding requirement applies.

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Upon removal of the safety pin, observe that the slider remains in the fully safe position. Should the fuze arm upon removal of the safety pin, it shall be rejected as a critical failure and forced to function before removal from the fixture in a manner as described in the end of this test description.

If the slider remains in the safe position upon removal of the safety pin, shear wire shall be removed. The fuze shall then be allowed to arm by raising the arming pin.

With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree. The magnitude of the impulse shall be 180 $\pm$ 20 gravities. Duration of the impulse shall be 1.5 $\pm$ 0.5 miliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact.

The trigger shall not release the detent ball and firing pin as a result of the shock impulse. Any fuze that functions during the No-Go test shall be classified as a critical failure.

Fuzes that pass the No-Go test shall be subjected to the arming pin/slider lock test by moving the slider back into its fully safe position and lowering the arming pin until it is fully seated. A new shear wire shall be installed. At this time observe that the slider remains in the fully safe position when unobstructed. Should arming occur, the fuze shall be rejected as a critical failure and forced to function before removal from the fixture in a manner as described in the end of this test description.

Fuzes that pass the arming pin/slider test shall be resafed by reinsertion of the safety pin. Resafing is considered complete at the end of installation of both the shear wire and safety pin.

Any fuze that cannot be properly resafed shall be rearmed in the fixture and forced to function before removal from the fixture. This can be accomplished by increasing the magnitude of the impulse to approximately 1,000 gravities.

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Should the armed fuze still not trigger, a hang-fire has occurred and must be handled accordingly. Provision must be made that will allow for removal of the failed fuze from the test equipment while maintaining a shielded condition."

PAGE 82

4.5.16, delete in its entirety and substitute:

"4.5.16 Fuzing go test. A random sample of fuzes shall be selected from each production lot and subjected to the Fuzing Go Test.

WARNING: This test is dangerous and all necessary safety precautions shall be taken. Test operators shall be adequately shielded from the fuzes and their holding fixtures during arming, application of impulse, and re-safing. Test equipment design and detailed operating and testing procedures must be approved by ARDC Safety Office.

The fuze shall be secured in a holding fixture in such a manner that the shock impulse that the fuze senses will not be attenuated or magnified beyond that applied. The design of the fixture shall be such as to not support the fuze beneath the barrier plate except in the area of the fuze seat.

After mounting the fuze to the holding fixture, but prior to performing the test, the safety pin and shear wire shall be removed. Removal of either the safety pin or shear wire shall be considered as beginning of the arming operation and thus safety shielding requirement applies.

Upon removal of the safety pin, observe that the slider remains in the fully safe (out-of-line) position. Should the fuze arm upon removal of the safety pin, the fuze shall be resafed and rejected as a critical arming failure.

If the slider remains in the safe position upon removal of the safety pin, the shear wire shall be removed. The fuze shall then be allowed to arm by raising the arming pin to the end of its stroke at which time the spring lock will engage.

With the fuze in the armed state, an impulse shall be applied to the fuze/fixture in a direction parallel to the axis of the fuze within one degree. The magnitude of the

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impulse shall be 680 ± 20 gravities. Duration of the impulse shall be 1.0 ± 0.5 milliseconds. The orientation of the fuze with respect to the impulse shall be such as to simulate a normal target impact.

The trigger shall release the detent ball and firing pin causing initiation of the explosive train. Any fuze that fails to function shall be considered a failure. A single failure shall cause the lot to the rejected.

Any fuze that fails the Fuzing Go Test shall be forced to function before removal from the fixture. This can be accomplished by increasing the magnitude of the impulse to approximately 1,000 gravities. Should the armed fuze still not trigger, a hang-fire has occurred and must be handled accordingly. Provision must be made that will allow for removal of the failed fuze from the test equipment while maintaining a shielded condition."

PAGES 21, 22, 26, 27, 34, 35, 36, 37, 38, 62, and 63

* Add: "9333825/9345168" in Next Higher Assembly block.

PAGES 23, 24, 25, and 29

Add: "9347185" in Next Higher Assembly block.

*The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletion) from the previous amendment were made. This was done as a convenienve only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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